FOVL Assignment

Objective: Develop and evaluate multilingual word embedding models that learn meaningful word representations across Indian languages.

Training Dataset

Use only the IndicCorpV2 dataset for training your models, available at:
`https://huggingface.co/datasets/rony000013/indiccorpV2-5k`

You may train separate monolingual embedding models for individual languages or a combined multilingual embedding model using the corpus of multiple languages. Your models must cover all languages included in the evaluation benchmark.

Tasks

1. **Model Development:**

   Implement and train word embedding models of your choice, such as GloVe, Word2Vec (CBOW or Skip-gram), FastText, etc.

   - Train all embeddings from scratch using only IndicCorpV2.
   - The embedding dimension must be exactly 300.
   - Pre-trained word embeddings and external training corpora are strictly prohibited.
   - You may use standard libraries and tools for implementing and training your models.

2. **Intrinsic Evaluation – Word Similarity:**

   Evaluate the learned word embeddings using the **Word Similarity Datasets for Indian Languages** released by Akhtar et al. (2017).

   Dataset repository:

   `https://github.com/syedsarfarazakhtar/Word-Similarity-Datasets-for-Indian-Languages`

   Evaluate the embeddings on **all languages for which word similarity datasets are available in the repository**.

   For each language:

   - Load the word pairs and their corresponding human-annotated similarity scores.
   - Obtain the trained embeddings of both words in each pair.
   - **OOV Handling:** Your model must be capable of generating meaningful embeddings for out-of-vocabulary (OOV) words as well. You are expected to implement an appropriate OOV handling strategy, such as subword-based representations or another suitable approach. Simply discarding word pairs containing OOV words is not permitted.
   - Calculate cosine similarity between the corresponding word embeddings:

$$s_{\text{model}}(w_i, w_j) = \frac{\mathbf{v}_{w_i}^\top \mathbf{v}_{w_j}}{\|\mathbf{v}_{w_i}\| \|\mathbf{v}_{w_j}\|}$$

- Calculate Spearman's rank correlation between model-generated cosine similarities and human similarity scores:

$$\rho_\ell = \text{Spearman}(s_{\text{model}}, s_{\text{human}})$$

Report the Spearman correlation separately for every evaluated language.

Note: The publicly available word similarity datasets should be used for development and validation. Final evaluation and leaderboard ranking will be performed using a separate private test dataset.

Scoring

The final leaderboard will use the macro-average Spearman correlation across all evaluated languages:

$$S_{\text{macro}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \rho_\ell$$

where $\mathcal{L}$ denotes the set of languages included in the evaluation benchmark.

The submission achieving the highest valid macro-average Spearman correlation on the private test dataset will be ranked highest.

1 Submission

Submit a single ZIP file containing:

1. **Trained Word Embeddings:** Trained word embeddings for all words appearing in the publicly specified word-pair datasets, saved in `.npy` format.
   Each word pair must have its corresponding two word embeddings. The embedding dimension must be exactly 300. The array should have the shape $(N, 2, 300)$, where N is the number of word pairs in the corresponding evaluation dataset. Preserve the original word-pair order.
2. **Vocabulary Mapping:** Include the corresponding vocabulary file to map words to their embedding vectors.
3. **Experimental Report:** Submit the report in PDF format.
4. **Source Code:** Submit the complete source code in Jupyter Notebook format (`.ipynb` only; no `.py` files).
   The notebook must include the implementation required to load the trained model and generate embeddings for arbitrary input words, including OOV words, for the final private evaluation.

Report Sections (10–11 pt font):

1. Introduction
2. Methodology
3. t-SNE Visualization (100 word samples per language)
4. Results (cosine similarity and Spearman correlation on the specified public benchmark)
5. Experiments
6. Analysis (5–10 word-pair examples, including correctly and incorrectly predicted similarities)
7. Discussion (design choices and limitations)
8. References

Important:

- Report vocabulary coverage and OOV statistics for each language.
- Clearly describe the OOV handling strategy.

- Disclose all pre-trained models, external datasets, and external tools used.
- All word embeddings must be trained from scratch using only IndicCorpV2.
- Human-annotated similarity scores from the benchmark must not be used for training or supervised optimization of the embeddings.
- Ensure that the submitted notebook can reproduce the reported results and generate embeddings for unseen words during final evaluation.

2 Deadline

The deadline for submission is **13 November**.